

IoT Simulator Backend API Documentation

Base URLs

Device Simulator Service

<http://localhost:8000/api/v1>

Telemetry Service

<http://localhost:8001/api/v1>

1. Start Simulation

Endpoint

POST /simulation/start

Purpose

Starts telemetry generation for a device.

Request Body

```
{  
  "device_id": "temp001",  
  "device_type": "temperature_sensor",  
  "interval": 5  
}
```

Parameters

Field	Type	Description
device_id	String	Unique device identifier
device_type	String	projector, camera, microphone, speaker, temperature_sensor
interval	Integer	Telemetry generation interval (seconds)

Success Response

```
{  
  "message": "Simulation Started"  
}
```

2. Stop Simulation

Endpoint

POST /simulation/stop

Request Body

```
{  
  "device_id": "temp001"  
}
```

Success Response

```
{  
  "message": "Simulation Stopped"  
}
```

3. Get Running Simulations

Endpoint

GET /simulation/running

Success Response

```
{  
  "count": 2,  
  "running_devices": [  
    "temp001",  
    "speaker001"  
  ]  
}
```

4. Get Device Telemetry History

Endpoint

GET /telemetry/{device_id}

Example

GET /telemetry/temp001

Query Parameter

Parameter	Description
time_range	Time range for telemetry (-30m, -1h, -24h etc.)

5. Get Latest Device Telemetry

Endpoint

GET /telemetry/latest/{device_id}

Example

GET /telemetry/latest/temp001

Success Response

```
{
  "device_id": "temp001",
  "device_type": "temperature_sensor",
  "temperature": 28.7,
  "battery": 96.2,
  "timestamp": "2026-06-28T11:35:00Z"
}
```

6. Telemetry Health Check

Endpoint

GET /telemetry/health

Success Response

```
{  
  "service": "Telemetry Service",  
  "status": "Running"  
}
```

Supported Device Types

- temperature_sensor
 - speaker
 - camera
 - microphone
 - projector
-

Frontend Usage

Dashboard Cards

Use

GET /telemetry/latest/{device_id}

Examples:

- Current Temperature
 - Battery Level
 - Volume
 - Brightness
 - FPS
-

Recharts Graphs

Use

GET /telemetry/{device_id}

with

?time_range=-30m

or

?time_range=-1h

The response contains the historical telemetry data required for plotting time-series charts.

Running Devices

Use

GET /simulation/running

to retrieve the list of currently active simulated devices and populate the dashboard dynamically.